

Section 1: Metadata

to be filled by the student

1.1. Project Information to be filled by the student

Title: Library Management System	
Section: L5	Instructor: Sir Umer Tariq

1.2. Student(s) Information

Name: Ismail Nooruddin	ID: in09958
Section: L5	Batch: 2028

Name: Syed Faraz Ahmed Shah Fida Hussain	ID: sf10162
Section: L5	Batch: 2028

Name: Hassan Akhter	ID: ha10609
Section: L5	Batch: 2028

Submission guideline: Save your project proposal as a pdf file and rename as Section<Number>-Group<Number>.pdf

Section 2: The Project

to be filled by the student

1. Project Description: *Please provide a brief introduction of the project including its scope.*

For example, a “toy version of Daraz app”

This project is a Library Management System that helps librarians manage book inventory and check in/out books in case a typical barcode scanning system is not available/working in the library. It would allow the members to search books and also see their account profile which would show their currently borrowed books and books borrowed in the past.

2. Functional Requirements

This section describes each function/feature provided by your system.

Sample functional Requirements:

User Types:

User A Member

User B Librarian

Functional Requirement 1: Register and Login

The system shall allow members and librarian to register an account and login.

Functional Requirement 2: Search/Filter Books

The system shall allow a member to search books through their name, or filter books via author name or genre and know the availability of the book.

Functional Requirement 3: User View Profile

The system shall allow a member to view his profile which would show the books they have currently borrowed, as well as those borrowed in the past.

Functional Requirement 4: Manage Inventory

The system shall allow a librarian to manage book inventory by adding new books or updating the number of copies of the books currently available.

Functional Requirement 5: Monitor the Order Status

The system shall allow a librarian to check in/out a book against the member Id.

Note:

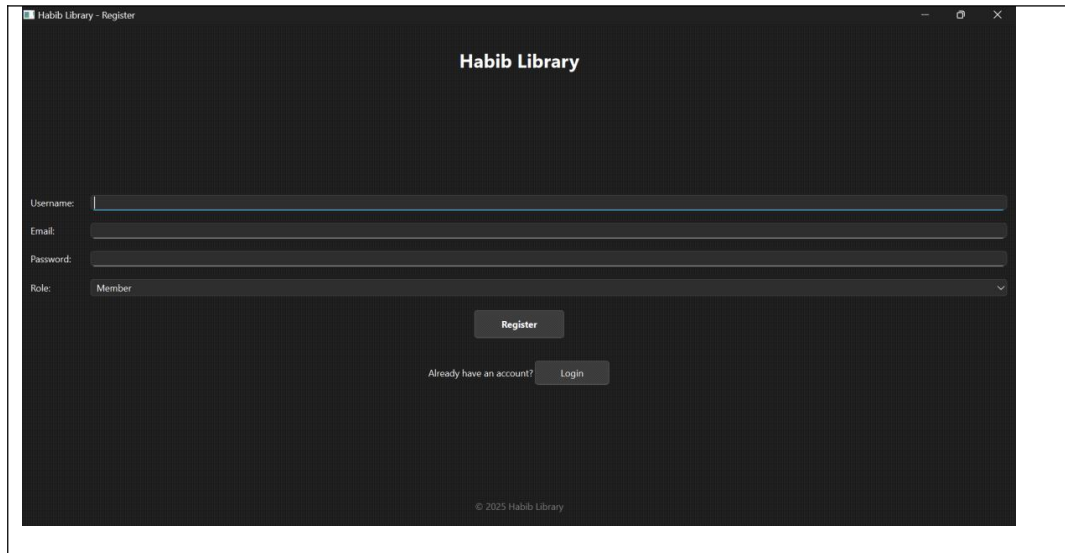
Functional requirements must contain at least one "interaction" between User Types

Buyer places an order -> Seller updates the status of an order -> Buyer sees the update status

3. UI Screens: *Provide images of all application screens, showcasing clear input and corresponding outputs. Ensure each image includes a concise caption explaining user action and expected/observed output. You can create these screens using Qt Designer.*

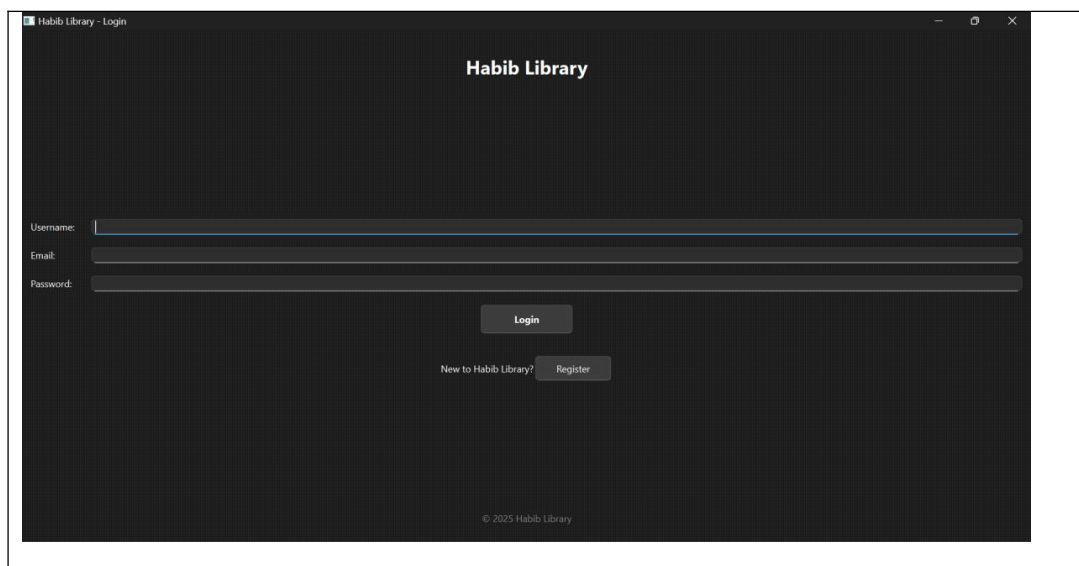
Functional Requirement 1: Register and Login

The system shall allow members and librarian to register an account and login.



The screenshot shows the 'Habib Library - Register' window. It features a dark-themed interface with the title 'Habib Library' at the top. Below the title, there are four input fields: 'Username:', 'Email:', 'Password:', and 'Role:'. The 'Role:' field is a dropdown menu currently set to 'Member'. Below these fields is a 'Register' button. Underneath the button, there is a link 'Already have an account?' followed by a 'Login' button. At the bottom center, there is a copyright notice '© 2025 Habib Library'.

UI Screen 1.A: Register screen

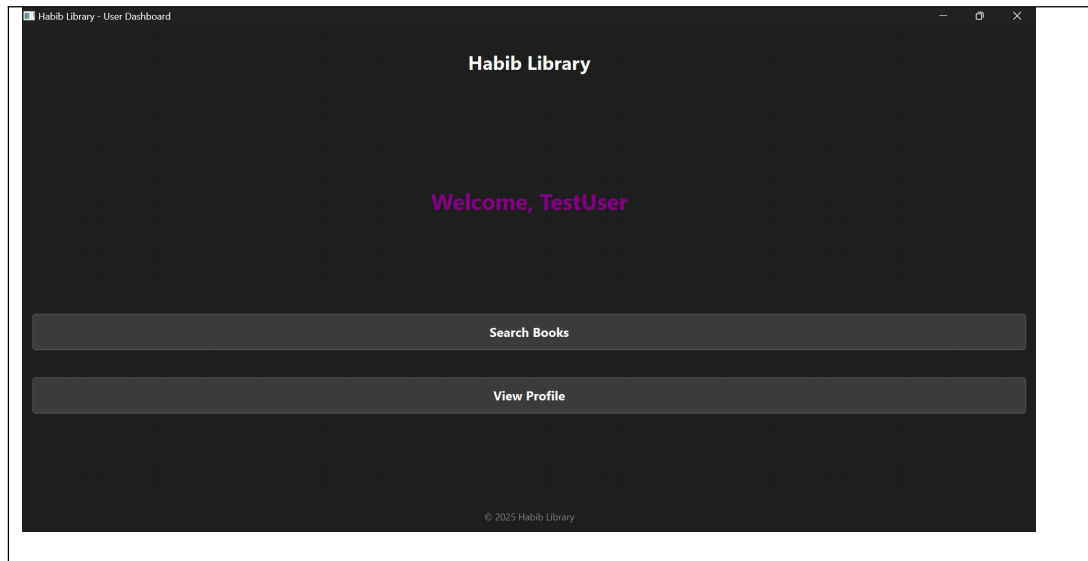


The screenshot shows the 'Habib Library - Login' window. It features a dark-themed interface with the title 'Habib Library' at the top. Below the title, there are three input fields: 'Username:', 'Email:', and 'Password:'. Below these fields is a 'Login' button. Underneath the button, there is a link 'New to Habib Library?' followed by a 'Register' button. At the bottom center, there is a copyright notice '© 2025 Habib Library'.

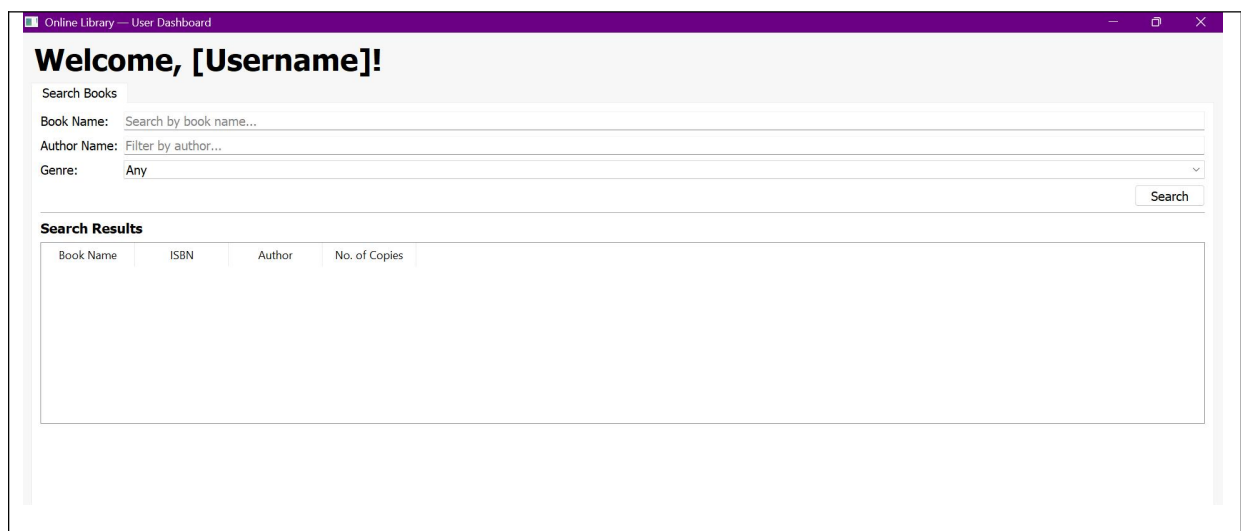
UI Screen 1.B: Log in screen

Functional Requirement 2: Search/Filter Books

The system shall allow a member to search books through their name, or filter books via author name or genre and know the availability of the book.



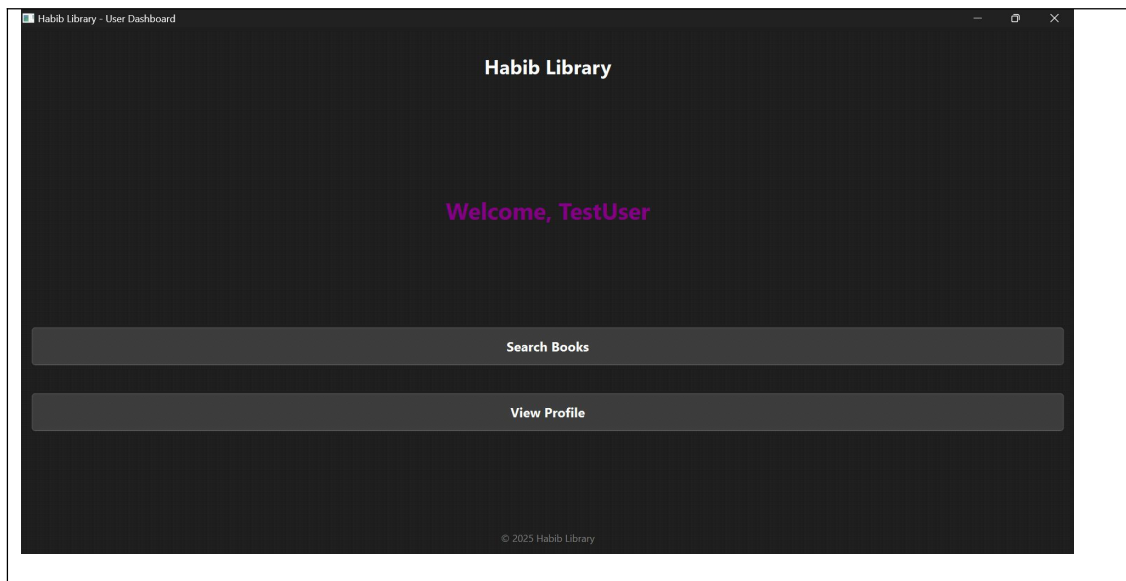
UI Screen 2.A: User Main Screen



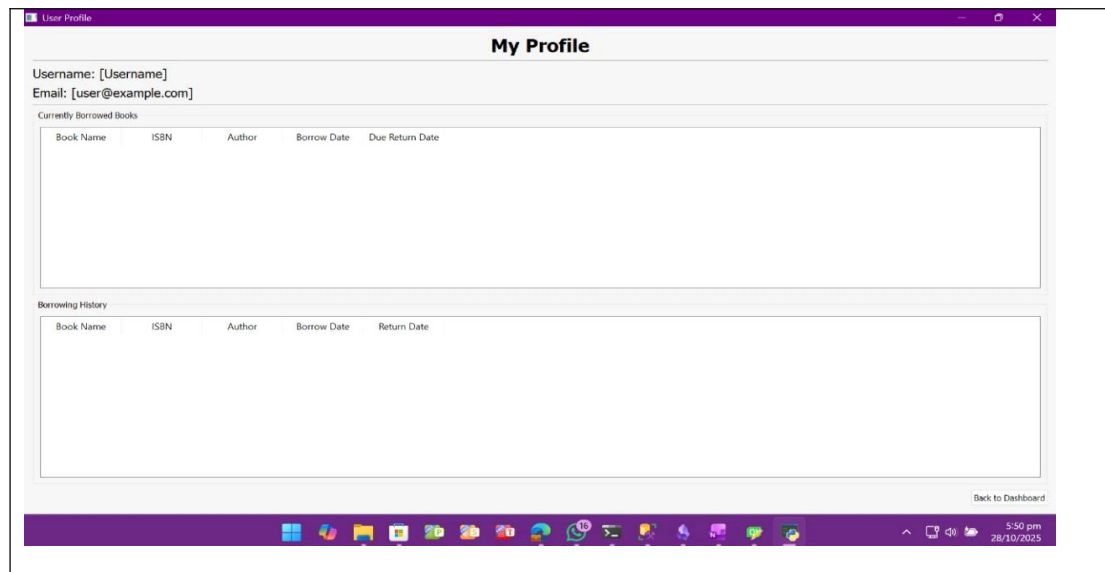
UI Screen 2.B: Search Books Screen

Functional Requirement 3: User View Profile

The system shall allow a member to view his profile which would show the books they have currently borrowed, as well as those borrowed in the past.



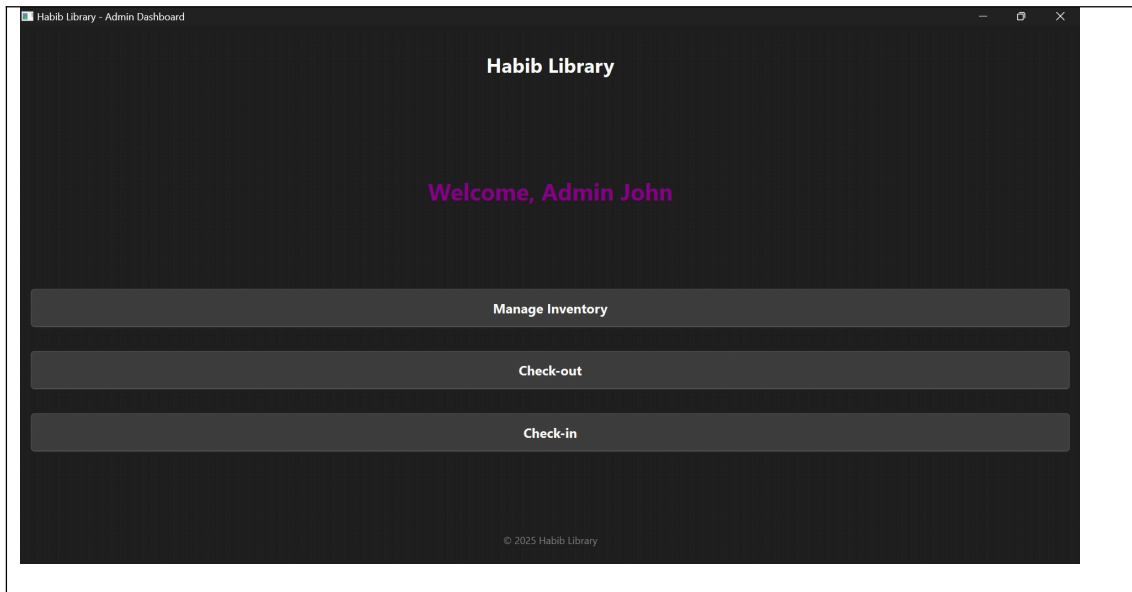
UI Screen 3.A: User Main Screen



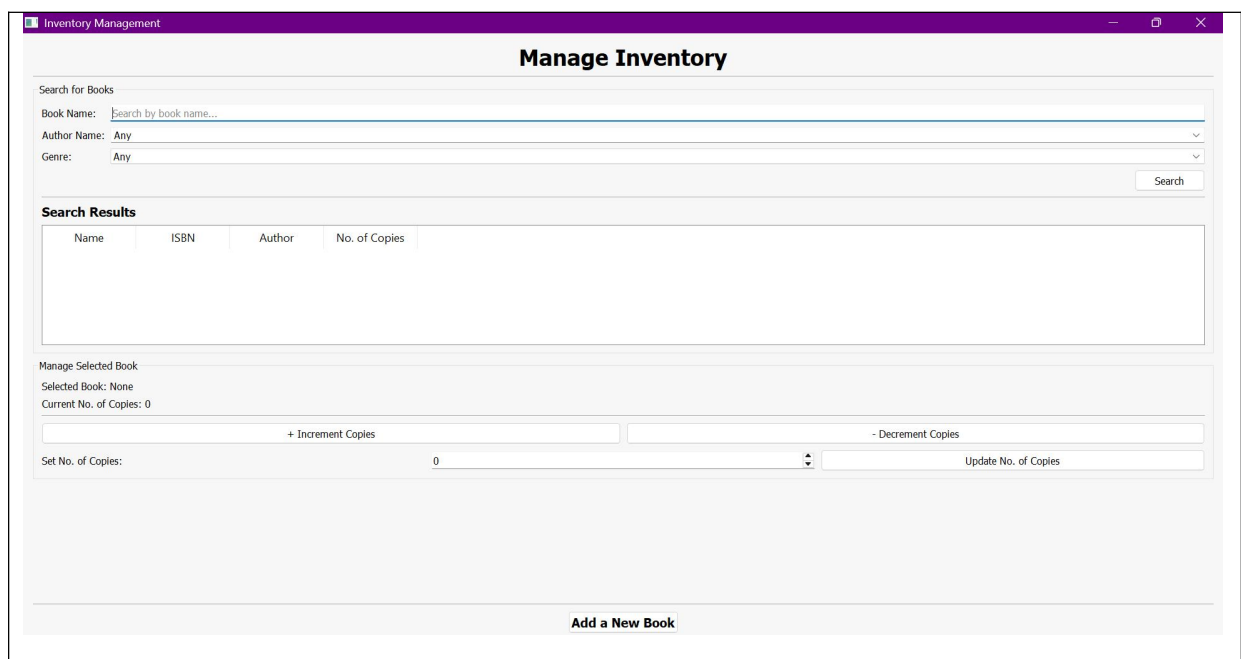
UI Screen 3.B: User Profile

Functional Requirement 4: Manage Inventory

The system shall allow a librarian to manage book inventory by adding new books or updating the number of copies of the books currently available.



UI Screen 4.A: Librarian Main Screen



UI Screen 4.B: Manage Current Books

Add New Book

Add New Book to Inventory

Book Information

Book Name: *

Enter book name...

ISBN: *

Enter ISBN (e.g., 978-3-16-148410-0)...

No. of Copies: *

1

Authors: *

Add new author...

+ Add

Genres: *

Fiction
Science Fiction
Mystery
Fantasy

* Required fields

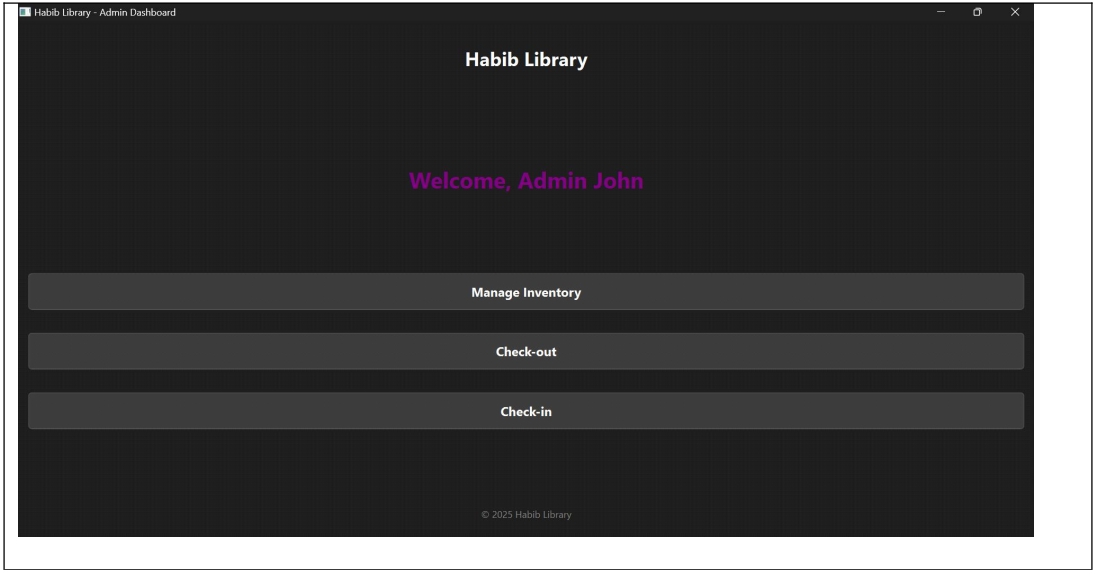
Cancel

Add Book

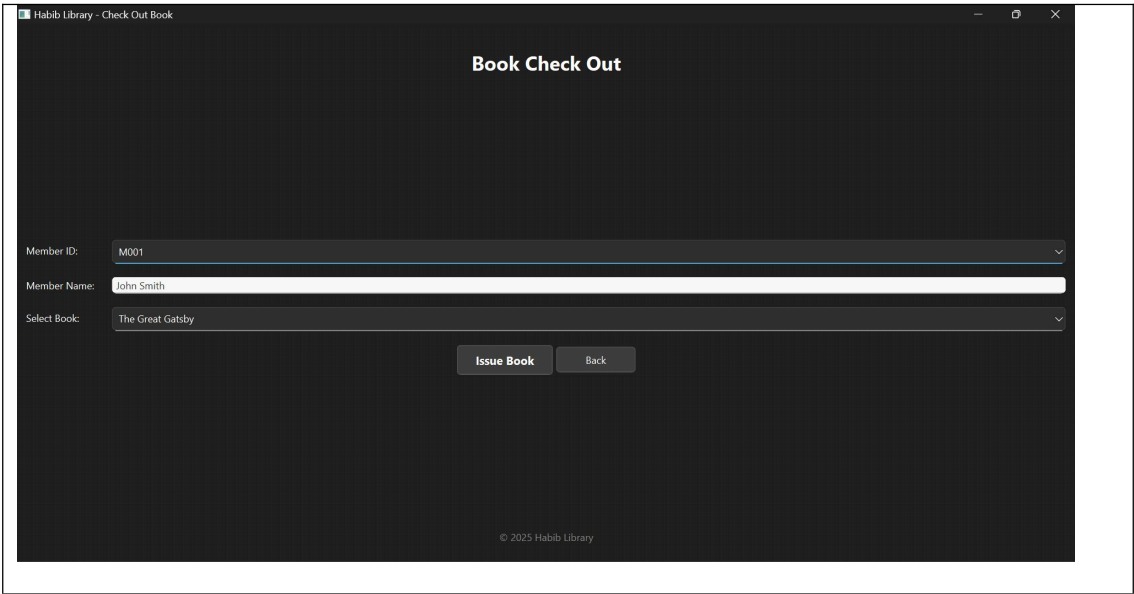
UI Screen 4.C: Add new Book

Functional Requirement 5: Monitor the Order Status

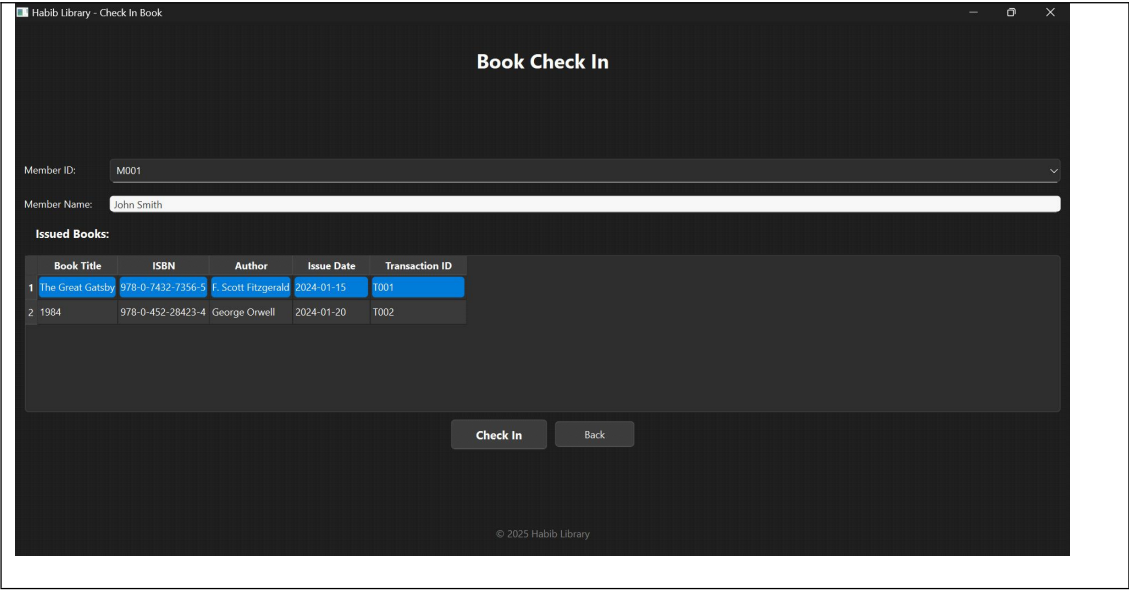
The system shall allow a librarian to check in/out a book against the member Id.



UI Screen 5.A: Librarian Main Screen

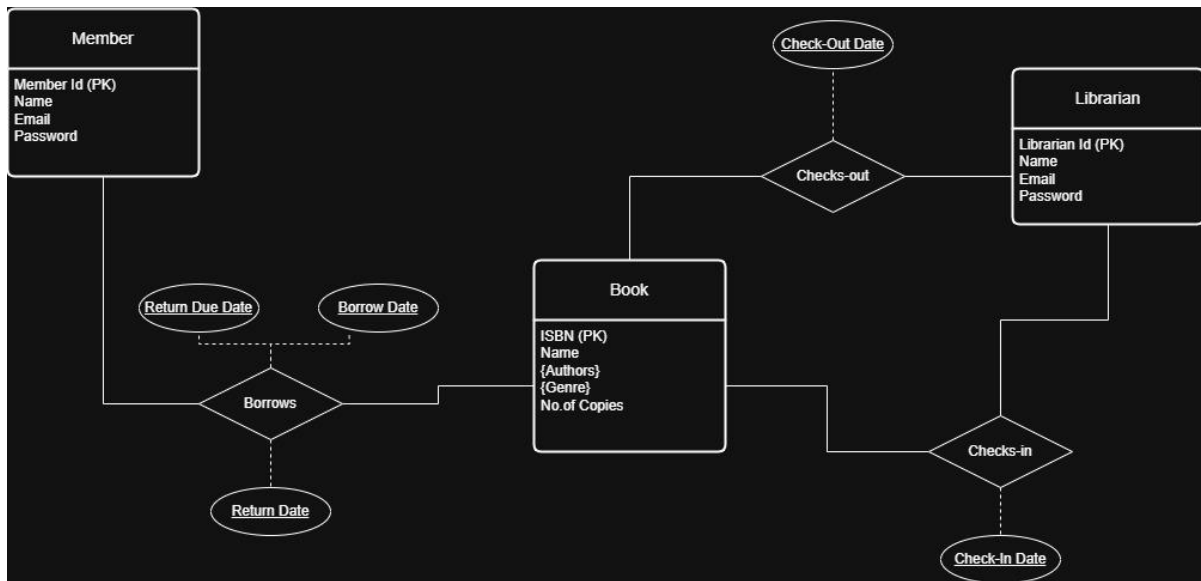


UI Screen 5.B: Check out screen

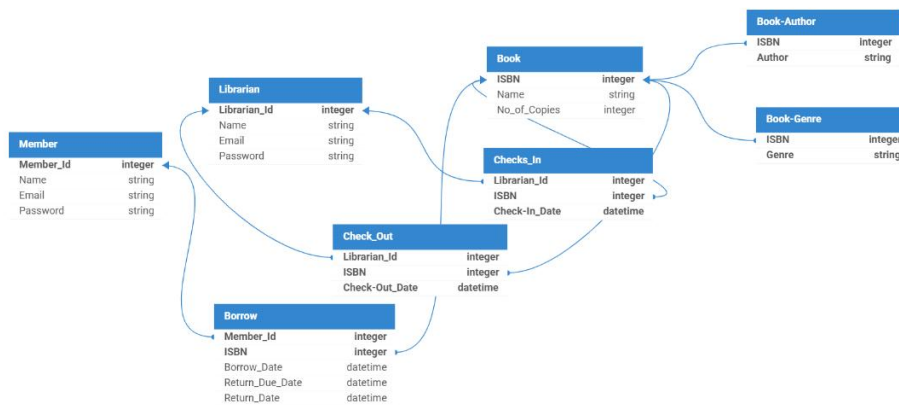


UI Screen 5.C: Check in

4. ERD



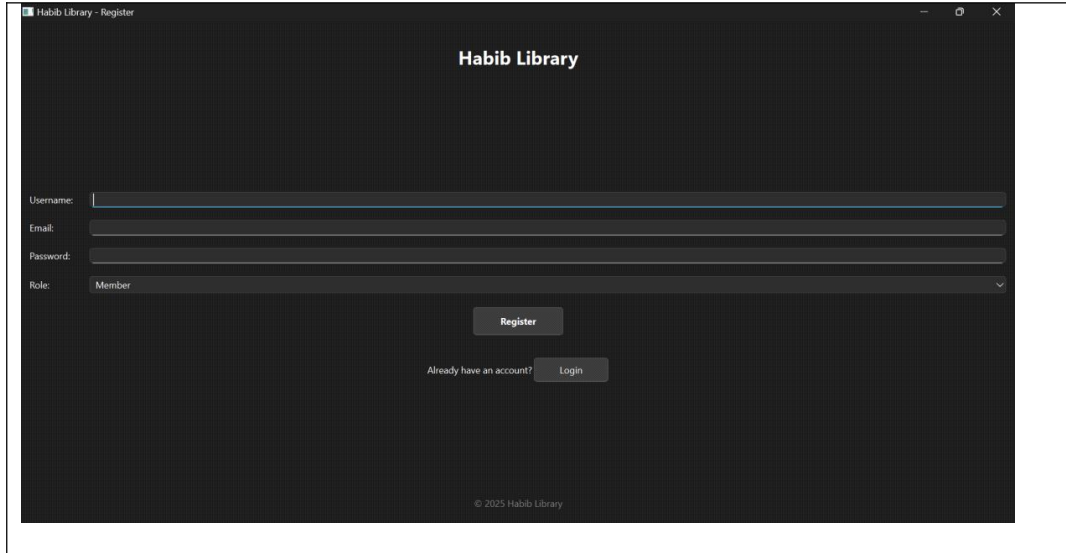
5. Relation Schema



6. SQL

Functional Requirement 1: Register and Login

The system shall allow members and librarian to register an account and login.



UI Screen 1.A: Register screen

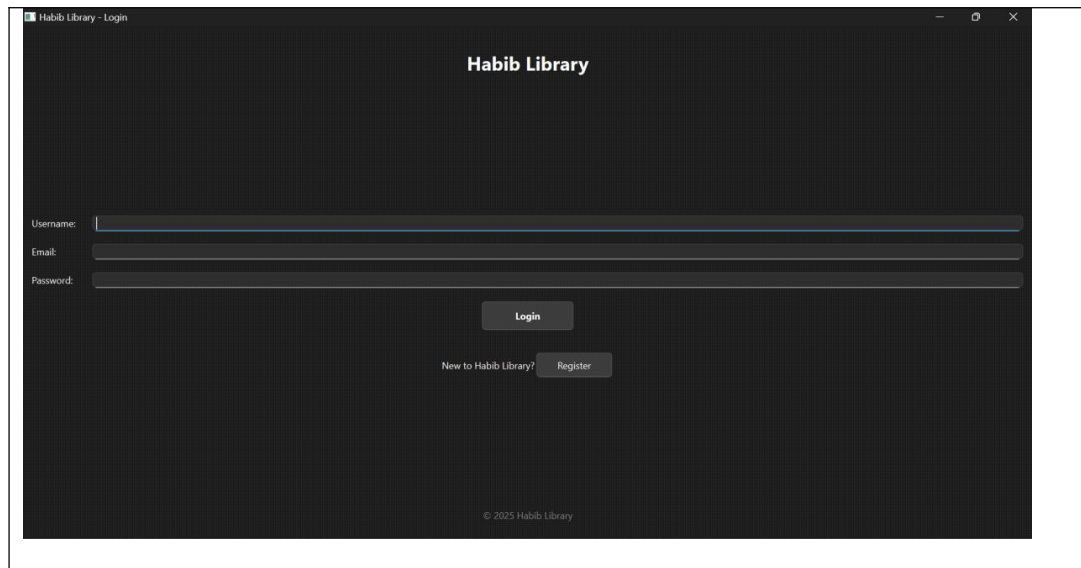
NOTE: The @ keyword/symbol represents input from the user.

SQL Query 1.A.1: (Registering a member)

```
INSERT INTO Member (Name, Email, Password)
OUTPUT INSERTED.Member_id
VALUES (@Name, @Email, @Password);
```

SQL Query 1.A.2: (Registering a librarian)

```
INSERT INTO Librarian (Name, Email, Password)
OUTPUT INSERTED.Librarian_id
VALUES (@Name, @Email, @Password);
```



UI Screen 1.B: Log in screen

SQL Query 1.B.1: (Checking for whether the user exists as a member or librarian in the database)

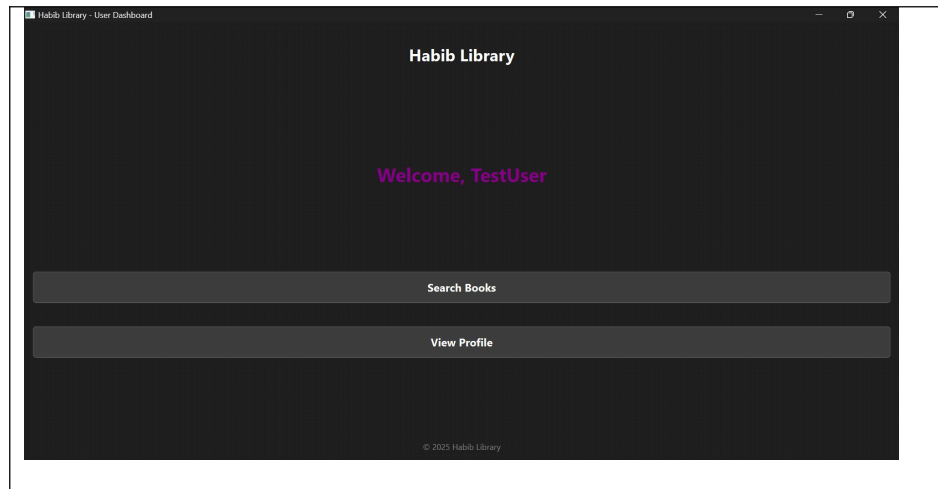
```
SELECT 'Member' as role, Member_id as user_id FROM Member
WHERE Name = @Name AND Email = @Email AND Password = @Password

UNION ALL

SELECT 'Librarian' as role, Librarian_id as user_id FROM Librarian
WHERE Name = @Name AND Email = @Email AND Password = @Password;
```

Functional Requirement 2: Search/Filter Books

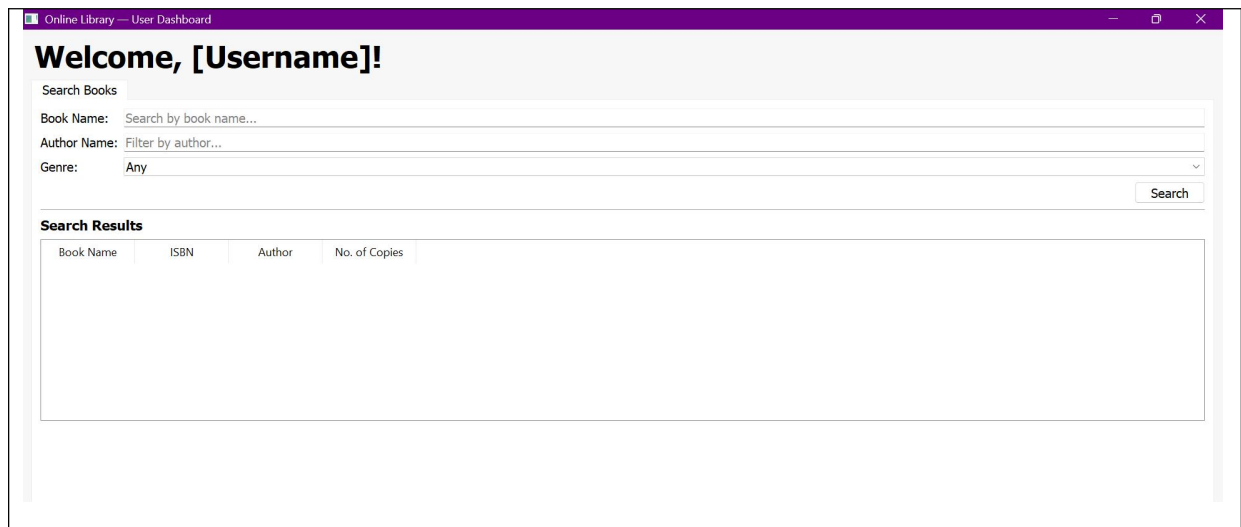
The system shall allow a member to search books through their name, or filter books via author name or genre and know the availability of the book.



UI Screen 2.A: User Main Screen

SQL Query 2.A.1:

No SQL query is needed for this screen as this is just a navigation screen with buttons to move to the user profile interface or the search books interface.



UI Screen 2.B: Search Book Screen

SQL Query 2.B.1(Searching The Book):

Instead of running separate queries for every possible combination of search filters (Name, Author, Genre), the system uses a single, optimized query. This query utilizes the STUFF and XML PATH functions to aggregate multiple authors and genres into single comma-separated strings (e.g., 'Fiction, Mystery'). This ensures that books with multiple authors do not appear as duplicate rows. The WHERE clause is constructed dynamically based on which fields the user fills out.

SELECT

b.ISBN,

b.Name AS BookName,

```

b.No_of_copies,

-- Subquery 1: Aggregates multiple Authors into a single comma-separated string
STUFF((

    SELECT ', ' + ba2.Author

    FROM Book_Author ba2

    WHERE ba2.ISBN = b.ISBN

    FOR XML PATH(""), TYPE).value('.', 'NVARCHAR(MAX)'), 1, 2, "

) AS Authors,

-- Subquery 2: Aggregates multiple Genres into a single comma-separated string
STUFF((

    SELECT ', ' + bg2.Genre

    FROM Book_Genre bg2

    WHERE bg2.ISBN = b.ISBN

    FOR XML PATH(""), TYPE).value('.', 'NVARCHAR(MAX)'), 1, 2, "

) AS Genres

FROM Book b

WHERE 1=1

-- The following conditions are appended dynamically based on user input:

-- 1. Filter by Book Name (if input provided)
-- AND b.Name LIKE '%@BookName%'

-- 2. Filter by Author (if input provided)
-- AND EXISTS (SELECT 1 FROM Book_Author ba WHERE ba.ISBN = b.ISBN AND ba.Author LIKE
'%@AuthorName%')

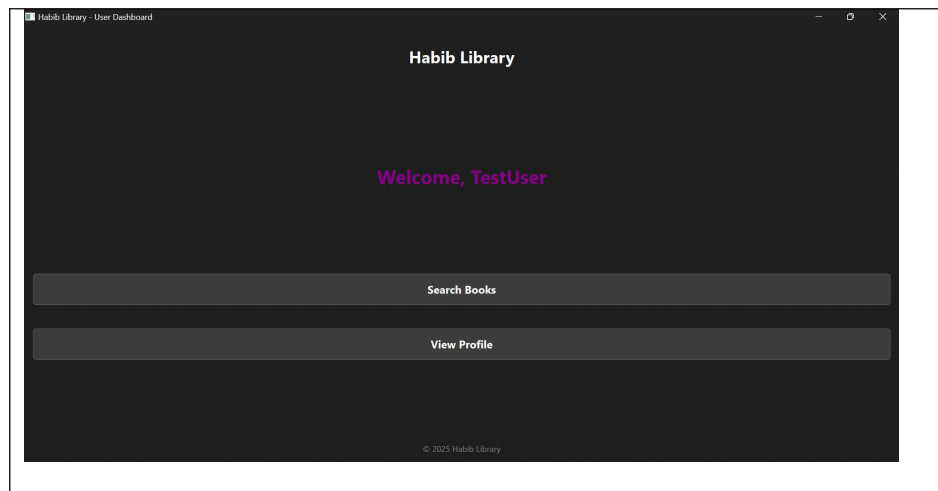
-- 3. Filter by Genre (if input provided)
-- AND EXISTS (SELECT 1 FROM Book_Genre bg WHERE bg.ISBN = b.ISBN AND bg.Genre =
'@SelectedGenre')

ORDER BY b.Name;

```

Functional Requirement 3: User View Profile

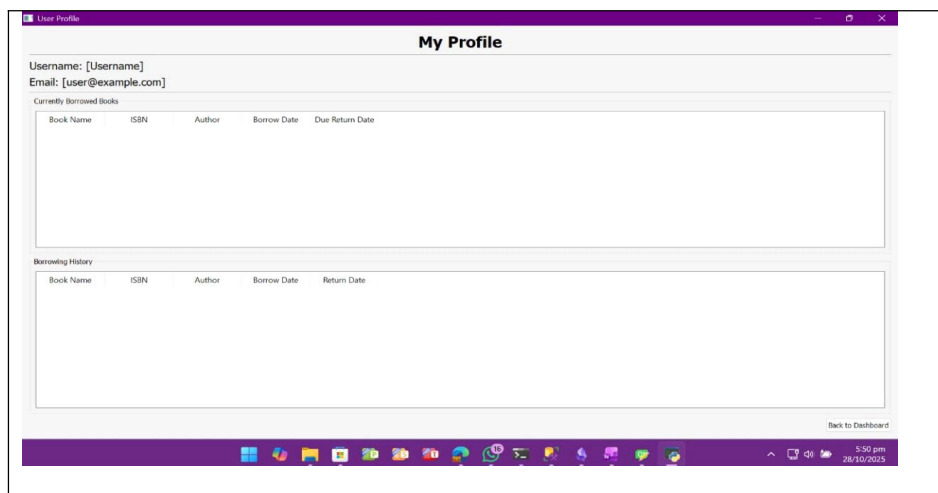
The system shall allow a member to view his profile which would show the books they have currently borrowed, as well as those borrowed in the past.



UI Screen 3.A: User Main Screen

SQL Query 3.A.1:

No SQL query is needed for this screen as this is just a navigation screen with buttons to move to the user profile interface or the search books interface.



UI Screen 3.B: User Profile

SQL Query 3.B: (To fetch the borrowed books; used the STUFF method again to ensure that if a borrowed book has 3 authors, the user still only sees 1 row in their history, not 3.)

SELECT

b.Name AS BookTitle,

bor.ISBN,

STUFF((

SELECT ', ' + ba2.Author

```
FROM Book_Author ba2

WHERE ba2.ISBN = b.ISBN

FOR XML PATH(""), TYPE).value('.', 'NVARCHAR(MAX)'), 1, 2, "

) AS Authors,

bor.Borrow_Date,

bor.Return_Due_Date

FROM Borrow bor

INNER JOIN Book b ON bor.ISBN = b.ISBN

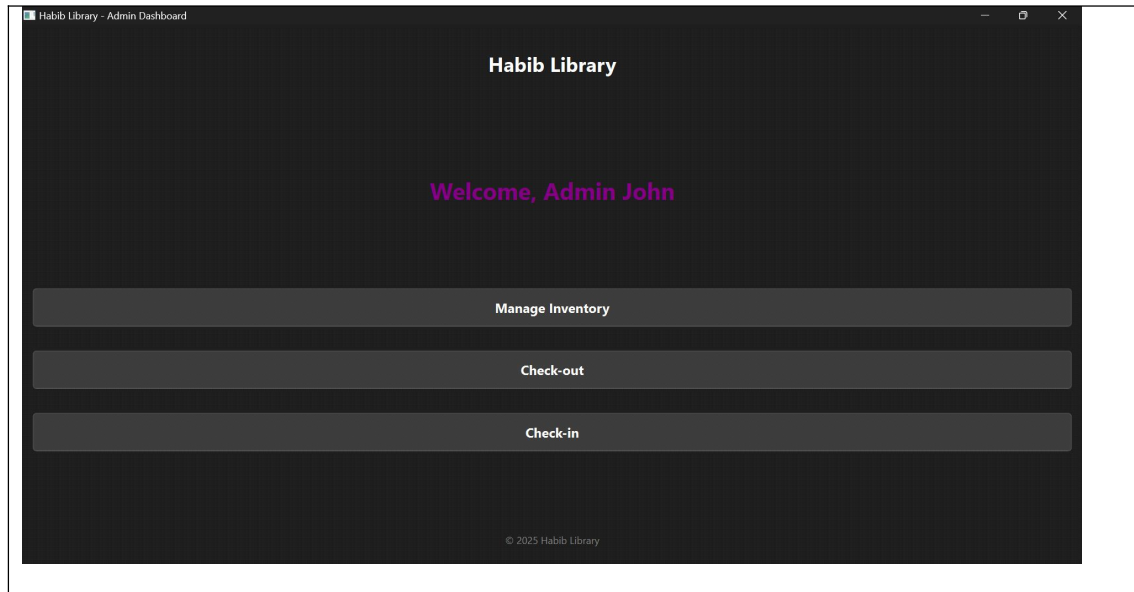
WHERE bor.Member_Id = @Member_Id

-- AND bor.Return_Date IS NULL (Included for "Currently Borrowed" section)

ORDER BY bor.Borrow_Date DESC;
```

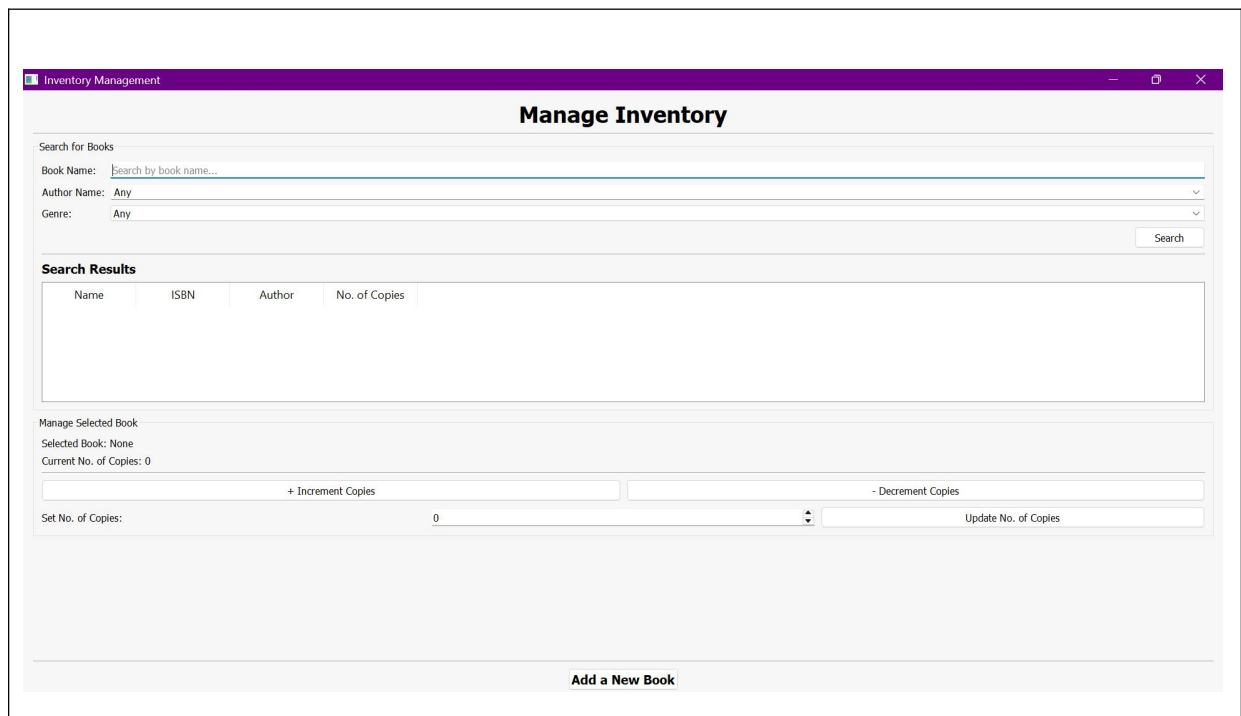
Functional Requirement 4: : Manage Inventory

The System Shall allow the librarian to mange inventory by adding books or updating the current books



UI Screen 4.A: Librarian Main Screen

No queries for this screen as this is just a navigation screen.



UI Screen 4.B: Mange Inventory

SQL Query 4.B.1(Searching and Incrementing Or Decrementing the No of Copies):

Note: For Searching the Book The same query will be used as of Screen of 2B.

UPDATE Book

SET No_of_copies = @NewCopies

WHERE ISBN = @ISBN;

-- Scenario 1 (Increment): Python calculates @NewCopies = (Current_Copies + 1)

-- Scenario 2 (Decrement): Python calculates @NewCopies = (Current_Copies - 1)

-- Scenario 3 (Set Value): Python passes the user input directly as @NewCopies

The screenshot shows a web application window titled 'Add New Book'. The main heading is 'Add New Book to Inventory'. Below this, there is a 'Book Information' section with the following fields: 'Book Name: *' with a text input field containing 'Enter book name...'; 'ISBN: *' with a text input field containing 'Enter ISBN (e.g., 978-3-16-148410-0)...'; and 'No. of Copies: *' with a numeric input field set to '1'. Below these is an 'Authors: *' section with a large text area and a '+ Add' button. Underneath is a 'Genres: *' section with a list box containing 'Fiction', 'Science Fiction', 'Mystery', and 'Fantasy'. At the bottom left of the form area, it says '* Required fields'. At the bottom right, there are two buttons: 'Cancel' and 'Add Book'.

SCREEN 4C: Add new book

SQL Query 4.C.1(Adding Book):

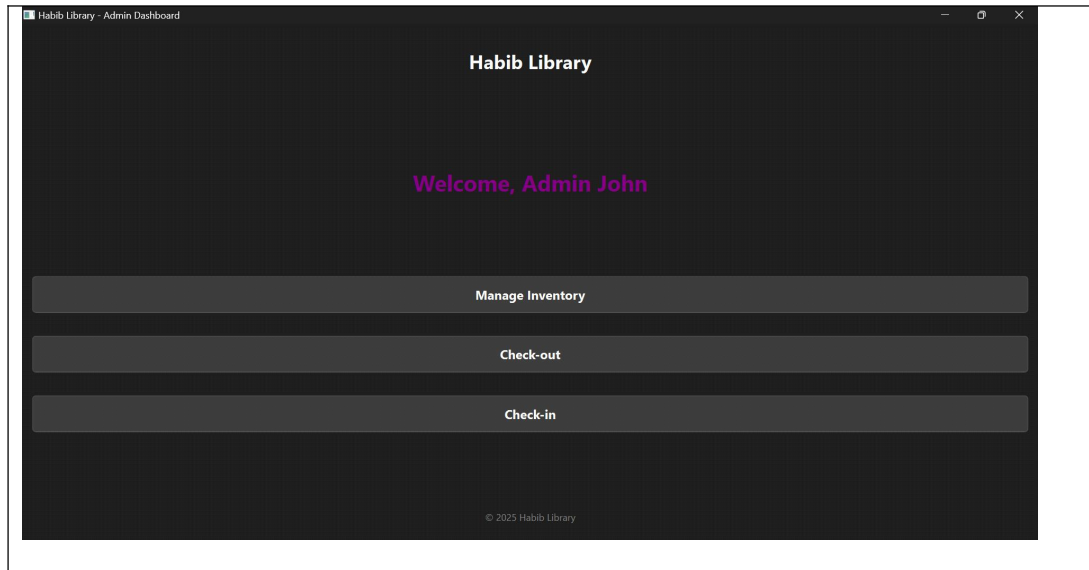
INSERT INTO Book (ISBN, Name, No_of_copies) VALUES (@ISBN, @Name, @Copies);

INSERT INTO Book_Author (ISBN, Author) VALUES (@ISBN, @Author); (Repeated for each author)

INSERT INTO Book_Genre (ISBN, Genre) VALUES (@ISBN, @Genre); (Repeated for each genre)

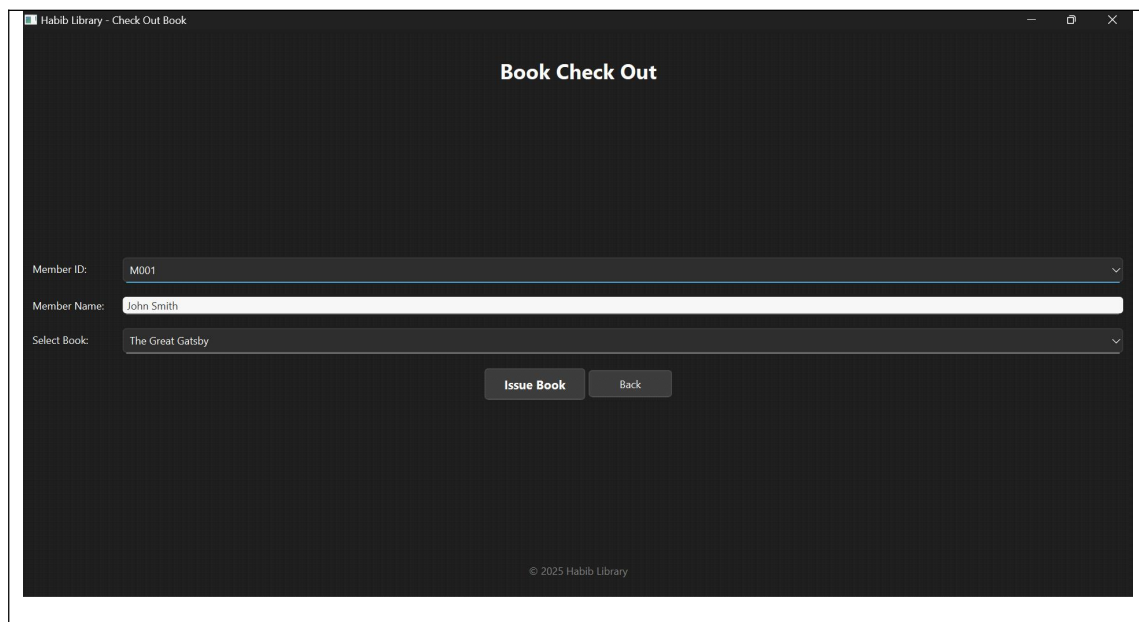
Functional Requirement 5: (Check out and Check In)

Allows the Librarian to check_In borrowed books and, checkout books that are being borrowed.



UI Screen 5.A: Librarian Main Screen

Again, no query as its just a navigation screen.



UI Screen 5.B: Check-out books

SQL Query 5.B.1(Check Out):

(Python handles the dates)

```
INSERT INTO Borrow (Member_Id, ISBN, Borrow_Date, Return_Due_Date, Return_Date)
```

```
VALUES (@MemID, @ISBN, @PyDate, @PyDueDate, NULL);
```

```
INSERT INTO Check_out (Librarian_Id, ISBN, Check_out_date)
```

```
VALUES (@LibID, @ISBN, @PyDate);
```

UPDATE Book SET No_of_copies = No_of_copies - 1 WHERE ISBN = @ISBN;

Habib Library - Check In Book

Book Check In

Member ID:

M001

Member Name:

John Smith

Issued Books:

	Book Title	ISBN	Author	Issue Date	Transaction ID
1	The Great Gatsby	978-0-7432-7356-5	F. Scott Fitzgerald	2024-01-15	T001
2	1984	978-0-452-28423-4	George Orwell	2024-01-20	T002

Check In

Back

© 2025 Habib Library

UI Screen 5.C: Check In Screen

SQL Query 5.C.1:

UPDATE Borrow

SET Return_Date = @PyDate

WHERE Member_Id = @MemID AND ISBN = @ISBN AND Return_Date IS NULL;

INSERT INTO Check_in (Librarian_Id, ISBN, Check_in_date)

VALUES (@LibID, @ISBN, @PyDate);

UPDATE Book SET No_of_copies = No_of_copies + 1 WHERE ISBN = @ISBN;

7. Final Demo Screenshots

Functional Requirement 1: Register and Login

The system shall allow members and librarian to register an account and login.

The image displays two screenshots of the Habib Library registration and login interface. Both screenshots feature a dark background with white text and form elements.

Top Screenshot (Member Registration):

- Title:** Habib Library
- Form Fields:**
 - Name: TestMember
 - Email: tm@gmail.com
 - Password: [masked]
 - Role: Member (dropdown menu)
- Buttons:** Register, Login
- Footer:** © 2025 Habib Library

Bottom Screenshot (Librarian Registration):

- Title:** Habib Library
- Form Fields:**
 - Name: TestLibrarian
 - Email: tl@gmail.com
 - Password: [masked]
 - Role: Librarian (dropdown menu)
- Buttons:** Register, Login
- Footer:** © 2025 Habib Library

Demo Screenshot 1.A: Register screen

1 SELECT * FROM Member

2 SELECT * FROM Librarian

89 %

✔ No issues found

Results

Messages

	Member_id	Name	Email	Password
1	1	TestMember	tm@gmail.com	testm

	Librarian_id	Name	Email	Password
1	1	TestLibrarian	tl@gmail.com	testl

Demo Screenshot 1.A: Result reflected in database

Habib Library

Name:TestMember

Email:tm@gmail.com

Password:•••••

Login

New to Habib Library?

Register

© 2025 Habib Library

Habib Library

Name:TestLibrarian

Email:tl@gmail.com

Password:•••••

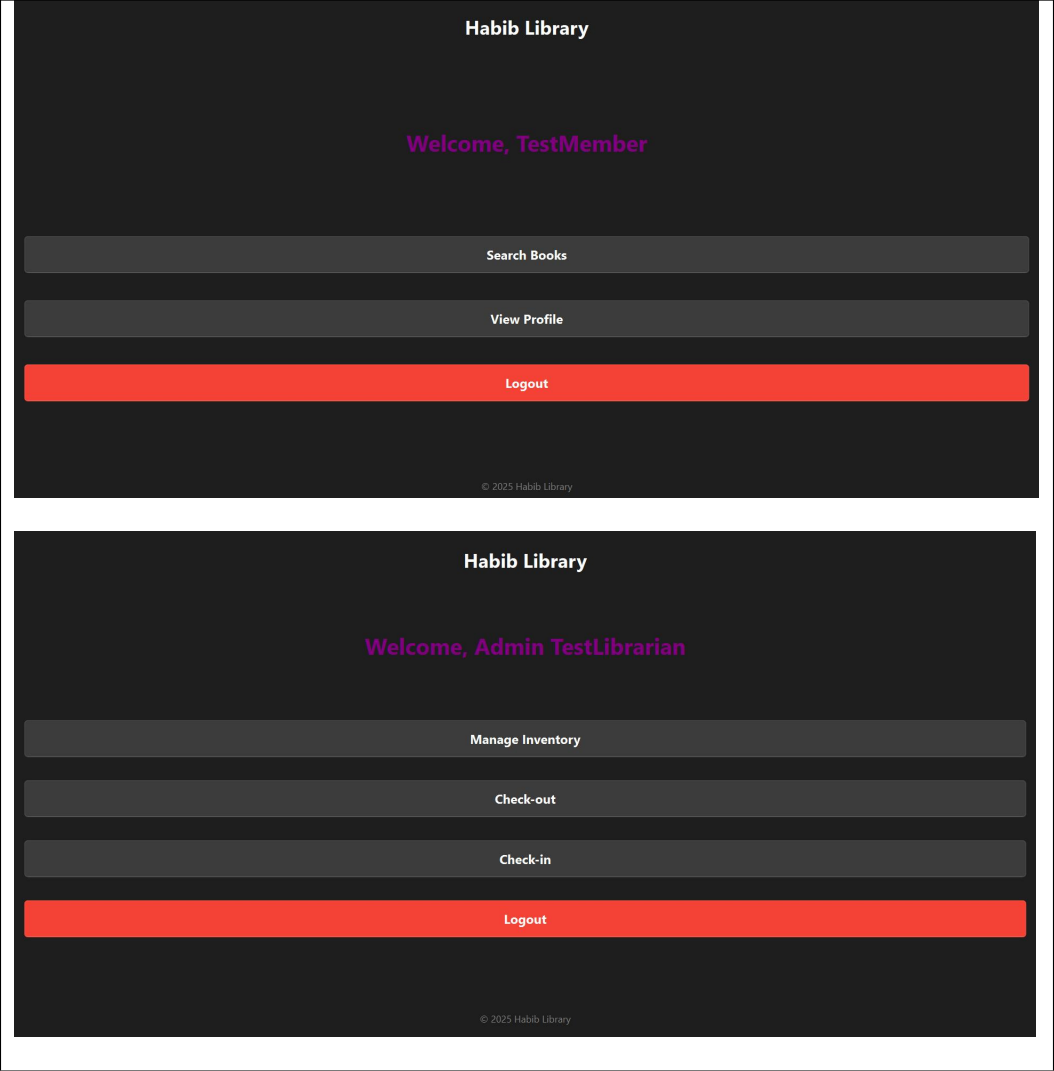
Login

New to Habib Library?

Register

© 2025 Habib Library

Demo Screenshot 1.B: Login screen



Demo Screenshot 1.B: Result reflected, Clicking Login bring you to your relevant dashboard

Functional Requirement 2: Search/Filter Books

The system shall allow a member to search books through their name, or filter books via author name or genre and know the availability of the book.

Searching without filling any fields(gives all books):

Welcome, TestMember!

- Search for Books

Book Name:

Author Name:

Genre:

Search Results (4 books found)

	Name	ISBN	Author	Genre	No. of Copies
1	Book A	123-ABC-567-X	Author A, Author B	Fiction, Mystery, Thriller	View Details
2	Book B	456-DEF-789-Y	Author B, Author C	Biography, History	View Details
3	Book C	012-GHI-013-Z	Author C, Author D	Mystery, Non-Fiction, Romance, Thriller	View Details
4	Book D	246-IJK-123-W	Author D	Fantasy	View Details

Searching only by name(gives all books in which the input string appears):

Welcome, TestMember!

- Search for Books

Book Name:

Author Name:

Genre:

Search Results (1 books found)

	Name	ISBN	Author	Genre	No. of Copies
1	Book A	123-ABC-567-X	Author A, Author B	Fiction, Mystery, Thriller	View Details

Welcome, TestMember!

- Search for Books

Book Name:

Author Name:

Genre:

Search Results (4 books found)

	Name	ISBN	Author	Genre	No. of Copies
1	Book A	123-ABC-567-X	Author A, Author B	Fiction, Mystery, Thriller	View Details
2	Book B	456-DEF-789-Y	Author B, Author C	Biography, History	View Details
3	Book C	012-GHI-013-Z	Author C, Author D	Mystery, Non-Fiction, Romance, Thriller	View Details
4	Book D	246-IJK-123-W	Author D	Fantasy	View Details

Filtering by author:

Welcome, TestMember!

Search for Books

Book Name:

Author Name:

Genre:

Search

Search Results (2 books found)

	Name	ISBN	Author	Genre	No. of Copies
1	Book A	123-ABC-567-X	Author A, Author B	Fiction, Mystery, Thriller	View Details
2	Book B	456-DEF-789-Y	Author B, Author C	Biography, History	View Details

Back to Dashboard

Filtering by genre:

Welcome, TestMember!

Search for Books

Book Name:

Author Name:

Genre:

Search

Search Results (1 books found)

	Name	ISBN	Author	Genre	No. of Copies
1	Book D	246-IJK-123-W	Author D	Fantasy	View Details

Back to Dashboard

Filtering by combinations:

Welcome, TestMember!

Search for Books

Book Name:

Author Name:

Genre:

Search

Search Results (1 books found)

	Name	ISBN	Author	Genre	No. of Copies
1	Book C	012-GHI-013-Z	Author C, Author D	Mystery, Non-Fiction, Romance, Thriller	View Details

Back to Dashboard

Welcome, TestMember!

Search for Books

Book Name:

Author Name:

Genre:

Search

Search Results (0 books found)

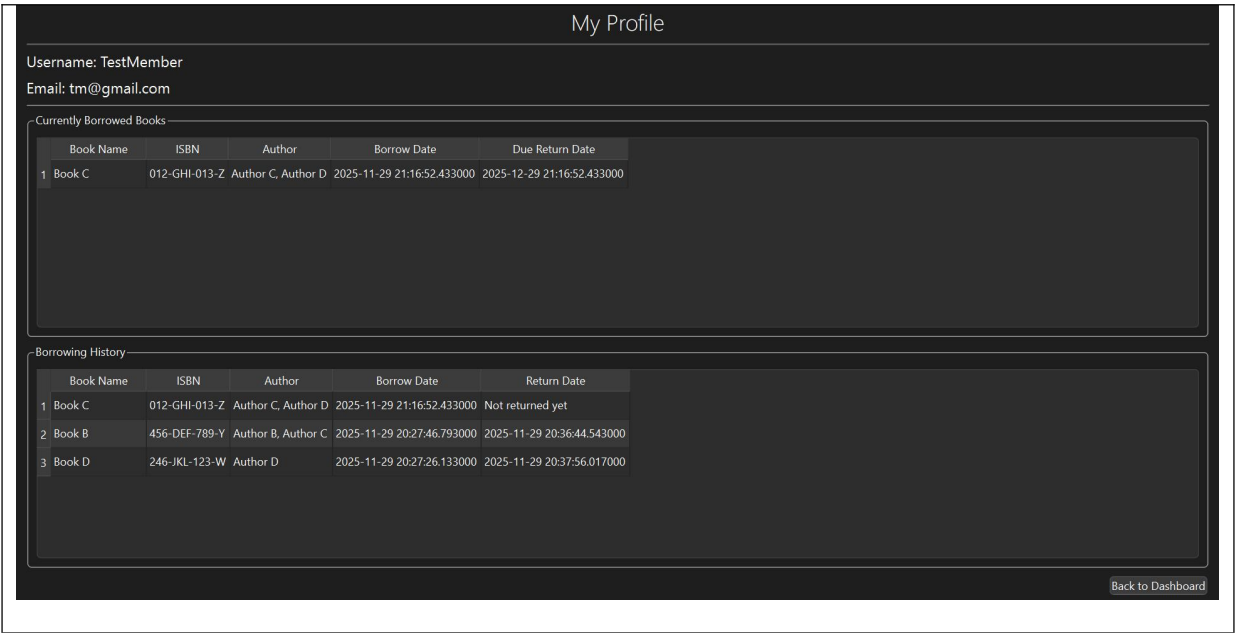
	Name	ISBN	Author	Genre	No. of Copies
1	No books...				

Back to Dashboard

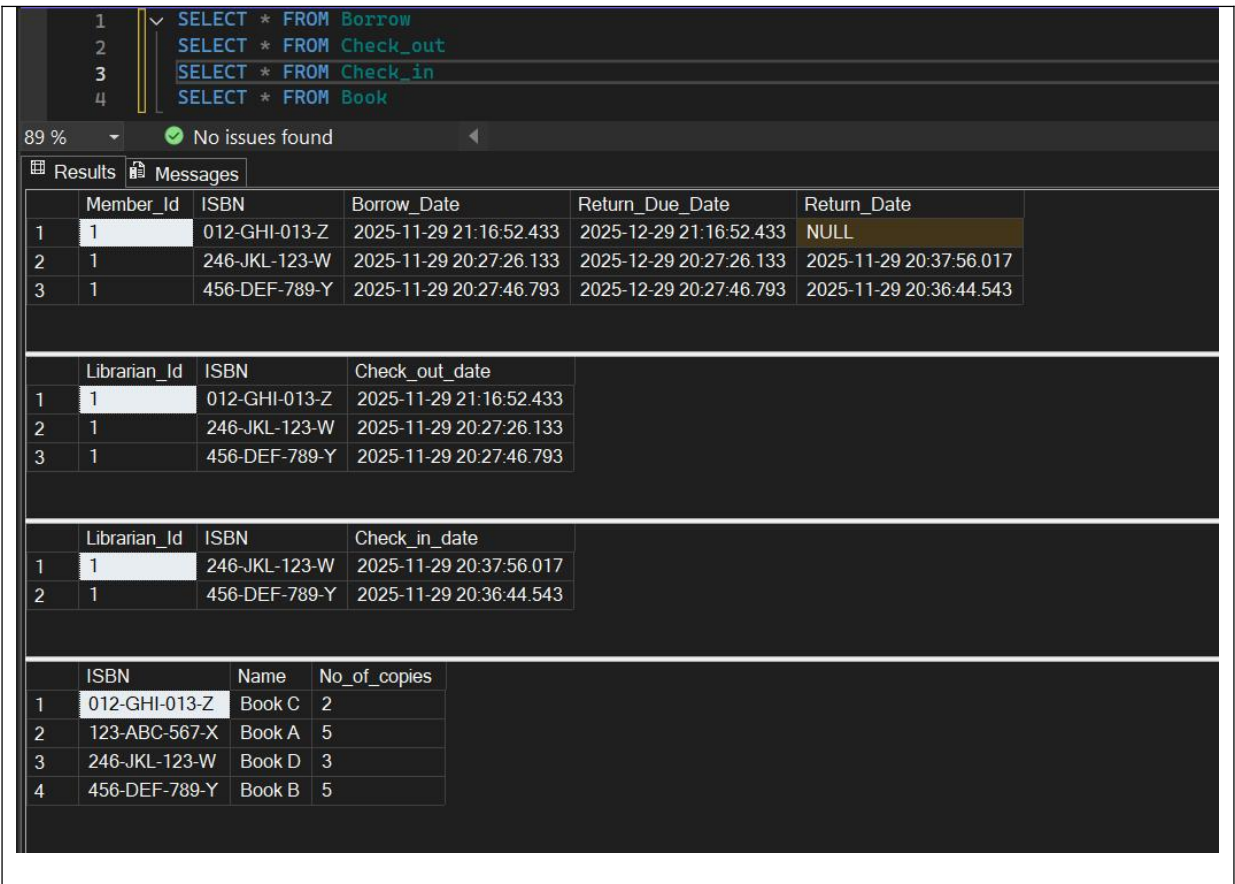
Demo Screenshot 2.A: Search/Filter Books

Functional Requirement 3: User View Profile

The system shall allow a member to view his profile which would show the books they have currently borrowed, as well as those borrowed in the past.



Demo Screenshot 3.A: User profile



Demo Screenshot 3.A: Records can be checked against the database

Functional Requirement 4: : Manage Inventory

The System Shall allow the librarian to mange inventory by adding books or updating the current books

Manage Inventory

Search for Books

Book Name:

Author Name:

Genre:

Search

Search Results

	Name	ISBN	Author	Genre	No. of Copies
1	Book A	123-ABC-567-X	Author A, Author B	Fiction, Mystery, Thriller	4
2	Book B	456-DEF-789-Y	Author B, Author C	Biography, History	6
3	Book C	012-GHI-013-Z	Author C, Author D	Mystery, Non-Fiction, Romance, Thriller	8
4	Book D	246-JKL-123-W	Author D	Fantasy	3

Manage Selected Book

Selected Book: Book A

Current No. of Copies: 4

+ Increment Copies

- Decrement Copies

Set No. of Copies: 4

Update No. of Copies

Add a New Book

Back to Dashboard

Manage Inventory

Search for Books

Book Name:

Author Name:

Genre:

Search

Search Results

	Name	ISBN	Author	Genre	No. of Copies
1	Book A	123-ABC-567-X	Author A, Author B	Fiction, Mystery, Thriller	5
2	Book B	456-DEF-789-Y	Author B, Author C	Biography, History	6
3	Book C	012-GHI-013-Z	Author C, Author D	Mystery, Non-Fiction, Romance, Thriller	8
4	Book D	246-JKL-123-W	Author D	Fantasy	3

Manage Selected Book

Selected Book: Book B

Current No. of Copies: 6

+ Increment Copies

- Decrement Copies

Set No. of Copies: 6

Update No. of Copies

Add a New Book

Back to Dashboard

Manage Inventory

Search for Books

Book Name:

Author Name:

Genre:

Search

Search Results

	Name	ISBN	Author	Genre	No. of Copies
1	Book A	123-ABC-567-X	Author A, Author B	Fiction, Mystery, Thriller	5
2	Book B	456-DEF-789-Y	Author B, Author C	Biography, History	5
3	Book C	012-GHI-013-Z	Author C, Author D	Mystery, Non-Fiction, Romance, Thriller	8
4	Book D	246-JKL-123-W	Author D	Fantasy	3

Manage Selected Book

Selected Book: Book C

Current No. of Copies: 8

+ Increment Copies

- Decrement Copies

Set No. of Copies: 3

Update No. of Copies

Add a New Book

Back to Dashboard

UI Screen 4.A: Mange Inventory(Incremented Book A copies, Decrementated Book B copies, and set Book C copies to 3)

Results

Messages

	ISBN	Name	No_of_copies
1	012-GHI-013-Z	Book C	8
2	123-ABC-567-X	Book A	4
3	246-JKL-135-W	Book D	3
4	456-DEF-789-Y	Book B	6

	ISBN	Author
1	012-GHI-013-Z	Author C
2	012-GHI-013-Z	Author D
3	123-ABC-567-X	Author A
4	123-ABC-567-X	Author B
5	246-JKL-135-W	Author D
6	456-DEF-789-Y	Author B
7	456-DEF-789-Y	Author C

	ISBN	Genre
1	012-GHI-013-Z	Mystery
2	012-GHI-013-Z	Non-Fiction
3	012-GHI-013-Z	Romance
4	012-GHI-013-Z	Thriller
5	123-ABC-567...	Fiction
6	123-ABC-567...	Mystery
7	123-ABC-567...	Thriller
8	246-JKL-135-W	Fantasy
9	456-DEF-789...	Biography
10	456-DEF-789...	History

Results

Messages

	ISBN	Name	No_of_copies
1	012-GHI-013-Z	Book C	3
2	123-ABC-567-X	Book A	5
3	246-JKL-123-W	Book D	3
4	456-DEF-789-Y	Book B	5

	ISBN	Author
1	012-GHI-013-Z	Author C
2	012-GHI-013-Z	Author D
3	123-ABC-567-X	Author A
4	123-ABC-567-X	Author B
5	246-JKL-123-W	Author D
6	456-DEF-789-Y	Author B
7	456-DEF-789-Y	Author C

	ISBN	Genre
1	012-GHI-013-Z	Mystery
2	012-GHI-013-Z	Non-Fiction
3	012-GHI-013-Z	Romance
4	012-GHI-013-Z	Thriller
5	123-ABC-567-X	Fiction
6	123-ABC-567-X	Mystery
7	123-ABC-567-X	Thriller
8	246-JKL-123-W	Fantasy
9	456-DEF-789-Y	Biography
10	456-DEF-789-Y	History

Demo Screenshot 4.A: Result reflected in database(First pic shows the old data, second pic shows the new data from after making the changes)

Add New Book to Inventory

Book Information

Book Name: *

Book A

ISBN: *

123-ABC-567-X

No. of Copies: *

4

Authors: *

Author A

Author B

Author C

Author D

Add new author...

+ Add

Genres: *

Fiction

Science Fiction

Mystery

Fantasy

Non-Fiction

* Required fields

Back

Add Book

Demo Screenshot 4.B: Adding Book

1

SELECT * FROM Book

2

SELECT * FROM Book_Author

3

SELECT * FROM Book_Genre

89 %

No issues found

Results

Messages

	ISBN	Name	No_of_copies
1	123-ABC-567-X	Book A	4

	ISBN	Author
1	123-ABC-567-X	Author A
2	123-ABC-567-X	Author B

	ISBN	Genre
1	123-ABC-567-X	Fiction
2	123-ABC-567-X	Mystery
3	123-ABC-567-X	Thriller

	ISBN	Name	No_of_copies
1	012-GHI-013-Z	Book C	8
2	123-ABC-567-X	Book A	4
3	246-JKL-135-W	Book D	3
4	456-DEF-789-Y	Book B	6

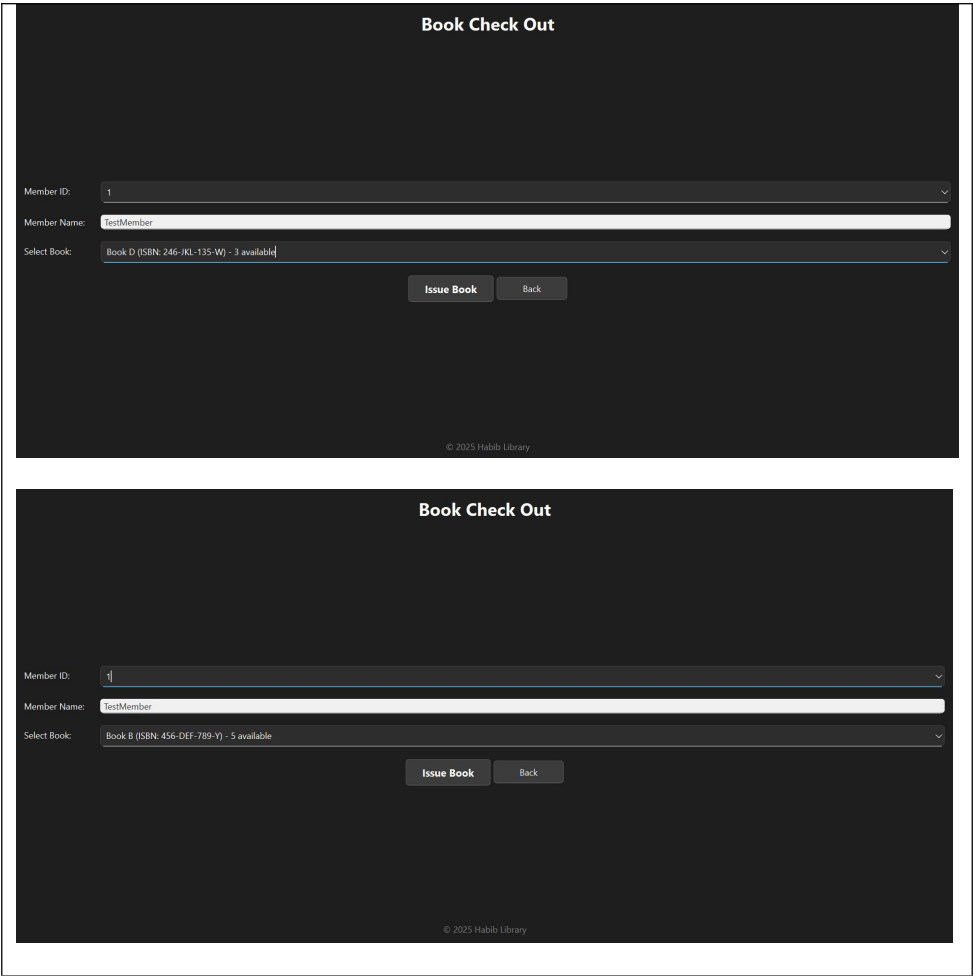
	ISBN	Author
1	012-GHI-013-Z	Author C
2	012-GHI-013-Z	Author D
3	123-ABC-567-X	Author A
4	123-ABC-567-X	Author B
5	246-JKL-135-W	Author D
6	456-DEF-789-Y	Author B
7	456-DEF-789-Y	Author C

	ISBN	Genre
1	012-GHI-013-Z	Mystery
2	012-GHI-013-Z	Non-Fiction
3	012-GHI-013-Z	Romance
4	012-GHI-013-Z	Thriller
5	123-ABC-567...	Fiction
6	123-ABC-567...	Mystery
7	123-ABC-567...	Thriller
8	246-JKL-135-W	Fantasy
9	456-DEF-789...	Biography
10	456-DEF-789...	History

Demo Screenshot 4.B: Result reflected in database(Added some more books as well)

Functional Requirement 5: (Check out and Check In)

Allows the Librarian to check_In borrowed books and, checkout books that are being borrowed.



Demo Screenshot 5.A: Checkout screen

1	SELECT * FROM Borrow
2	SELECT * FROM Check_out
3	SELECT * FROM Check_in
4	SELECT * FROM Book

89 %
No issues found

ResultsMessages

Member_Id	ISBN	Borrow_Date	Return_Due_Date	Return_Date
1	246-JKL-123-W	2025-11-29 20:27:26.133	2025-12-29 20:27:26.133	NULL
2	456-DEF-789-Y	2025-11-29 20:27:46.793	2025-12-29 20:27:46.793	NULL

Librarian_Id	ISBN	Check_out_date
1	246-JKL-123-W	2025-11-29 20:27:26.133
2	456-DEF-789-Y	2025-11-29 20:27:46.793

Librarian_Id	ISBN	Check_in_date

ISBN	Name	No_of_copies
012-GHI-013-Z	Book C	3
123-ABC-567-X	Book A	5
246-JKL-123-W	Book D	2
456-DEF-789-Y	Book B	4

Demo Screenshot 5.A: Result reflected in database(Check-in table and Return_Date in Borrow table are empty because these books haven't been returned yet, results after returning book are shown below. Plus no.of books are also decremented after checking out)

Book Check In

Member ID:

1

Member Name:

TestMember

Issued Books:

	Book Title	ISBN	Authors	Borrow Date	Due Date
1	Book B	456-DEF-789-Y	Author B, Author C	2025-11-29 20:27:46.793000	2025-12-29 20:27:46.793000
2	Book D	246-JKL-123-W	Author D	2025-11-29 20:27:26.133000	2025-12-29 20:27:26.133000

Check In

Back

© 2025 Habib Library

Book Check In

Member ID:

1

Member Name:

TestMember

Issued Books:

	Book Title	ISBN	Authors	Borrow Date	Due Date
1	Book D	246-JKL-123-W	Author D	2025-11-29 20:27:26.133000	2025-12-29 20:27:26.133000

Check In

Back

© 2025 Habib Library

Demo Screenshot 5.B: Checkin screen

1

2

3

4

SELECT * FROM Borrow

SELECT * FROM Check_out

SELECT * FROM Check_in

SELECT * FROM Book

89 %

No issues found

Results

Messages

	Member_Id	ISBN	Borrow_Date	Return_Due_Date	Return_Date
1	1	246-JKL-123-W	2025-11-29 20:27:26.133	2025-12-29 20:27:26.133	2025-11-29 20:37:56.017
2	1	456-DEF-789-Y	2025-11-29 20:27:46.793	2025-12-29 20:27:46.793	2025-11-29 20:36:44.543

	Librarian_Id	ISBN	Check_out_date
1	1	246-JKL-123-W	2025-11-29 20:27:26.133
2	1	456-DEF-789-Y	2025-11-29 20:27:46.793

	Librarian_Id	ISBN	Check_in_date
1	1	246-JKL-123-W	2025-11-29 20:37:56.017
2	1	456-DEF-789-Y	2025-11-29 20:36:44.543

	ISBN	Name	No_of_copies
1	012-GHI-013-Z	Book C	3
2	123-ABC-567-X	Book A	5
3	246-JKL-123-W	Book D	3
4	456-DEF-789-Y	Book B	5

Demo Screenshot 5.B: Result reflected in database(Both books checked in again. No.of copies for them also increments back to what they initially were)